

THE AI PERIODIC TABLE

Option B — Hybrid Pipeline + Ecosystem · 11 Chapters · Course Blueprint

This table is not merely a vocabulary list. It is a map of how modern AI systems move from raw data to real-world applications: data becomes structure, structure becomes meaning, meaning powers models, models interact through prompts and systems, and agentic workflows eventually act in the world.

Data Sources → Data Preparation → Representations → Model Learning → Model Families → Interaction Layer → System Integration → Agentic Design → Developer Frameworks → Automation Platforms → Applied AI Systems

KNOWLEDGE FOUNDATION · COLUMNS 1-3 INTELLIGENCE ENGINE · COLUMNS 4-5 INTERACTION & INTEGRATION · COLUMNS 6-7 AGENTIC BUILD LAYER · COLUMNS 8-10 APPLIED AI SYSTEMS · COLUMN 11

KNOWLEDGE FOUNDATION Data → Preparation → Representation			INTELLIGENCE ENGINE Learning → Model Families		INTERACTION & INTEGRATION Prompts → APIs / Tools		AGENTIC BUILD LAYER Agents → Frameworks → Platforms			APPLIED AI SYSTEMS Real-world use cases
I DATA SOURCES	II DATA PREPARATION	III REPRESENTATIONS	IV MODEL LEARNING	V MODEL FAMILIES	VI INTERACTION LAYER	VII SYSTEM INTEGRATION	VIII AGENTIC DESIGN	IX DEVELOPER FRAMEWORKS	X AUTOMATION PLATFORMS	XI APPLIED AI SYSTEMS
Td Training Data 1.1	Tk Tokenization 1.2	Em Embeddings 1.3	Tr Training 1.4	Gp GPT 1.5	Pr Prompt 1.6	Ap API 1.7	Ag Agent 1.8	Lc LangChain 1.9	N8 n8n 1.10	Rb Robotics 1.11
Sd Synthetic Data 2.1	Ch Chunking 2.2	Vs Vector Space 2.3	Ft Fine-Tuning 2.4	Cl Claude 2.5	Sp System Prompt 2.6	Ep Endpoint 2.7	G1 Goals 2.8	Li LlamaIndex 2.9	Zp Zapier 2.10	Es Embedded Systems 2.11
Dc Documents 3.1	Cn Cleaning 3.2	Vd Vector Database 3.3	R1 RLHF 3.4	Gm Gemini 3.5	Pe Prompt Engineering 3.6	Fn Function Calling 3.7	P1 Planning 3.8	Lg LangGraph 3.9	Mk Make 3.10	Dt Digital Twins 3.11
Ia Images / Audio 4.1	Ps Parsing 4.2	Ss Similarity Search 4.3	Ev Evaluation 4.4	Lm Llama 4.5	Ce Context Engineering 4.6	Mc MCP 4.7	Tu Tool Use 4.8	Ca CrewAI 4.9	Vf Voiceflow 4.10	De Digital Employees 4.11
Ed Enterprise Data 5.1	Nm Normalization 5.2	Rt Retrieval 5.3	A1 Alignment 5.4	Ms Mistral 5.5	Gd Guardrails 5.6	Sd SDK 5.7	Ma Multi-Agent 5.8	Au AutoGen 5.9	F1 Flowise 5.10	As AI Assistants 5.11
Sn Sensor Data 6.1		Rg RAG 6.3		Gr Granite 6.5	Ts Temperature 6.6	Wh Webhooks 6.7	Et Agent Ethics 6.8		Cr Cursor 6.10	Ao AI Operations 6.11

How to Read This Table

Columns = Chapters

Each column is a major instructional unit. Students move left to right from raw data to deployed AI applications.

Rows = Increasing System Maturity

Rows loosely move from foundational concepts toward more applied or specialized forms. Empty cells are intentional, not missing content.

Cells = Subchapters

Each cell can become one or more slides, mini-exercises, labs, or real-world stories depending on the course depth.

The Student Journey Through the Table

A student can trace a real build path across the table: **Training Data** → **Tokenization** → **Embeddings** → **Model Family** → **Prompt** → **API** → **Agent** → **LangChain** → **n8n** → **Robotics**. That path is the story of AI moving from information, to meaning, to action, and finally into the physical world.